

HEREDITY

- **Heredity** is the process of passing genetic traits(features) from one generation to the next. It's also known as genetics.
- Traits are the specific feature that can be observed in an individual and are influenced by both genetic factors and environmental interactions.
- **Inheritance:**
 - ✓ Inheritance is the pathway of genetic traits and how they are expressed from one generation to the next.
 - ✓ It's the pattern of heredity.
 - ✓ It determines an organism's characteristics, such as eye colour or height, and follows patterns like dominant and recessive traits.
 - ✓ Inheritance from the previous generation serves two main purposes:
 - ✚ **Common Basic Body Design:** It ensures that offspring retain the fundamental structure and features of the species (e.g., species-specific characteristics).
 - ✚ **Subtle Changes:** It introduces small variations in the traits of offspring, which are inherited from their parents and contribute to diversity within the species.
- **Variations:**
 - ✓ Variations refer to the differences in traits or characteristics between individuals within a species.
 - ✓ These differences can be physical (e.g., size, color) or behavioural and are caused by genetic differences, environmental factors, or a combination of both.
 - ✓ Types of Variations:
 - (i) Genetic Variation:**
 - Differences in DNA that lead to variations in traits.
 - This can occur through mutations, genetic recombination during sexual reproduction, or gene flow between populations.
 - **Mutation:**
 - ❑ A mutation is a change in the DNA sequence of an organism, virus, or extrachromosomal DNA.
 - ❑ Mutations can be harmful, beneficial, or have no effect.
 - ❑ Mutations can be caused by:
 - ❑ Mistakes during cell division
 - ❑ Exposure to DNA-damaging agents in the environment, such as chemicals and radiation
 - ❑ A viral infection
 - **Genetic Recombination:**
 - ❑ During sexual reproduction, offspring inherit genetic material from two parents, leading to new combinations of genes.
 - ❑ This creates greater diversity in the offspring, with individuals showing more noticeable differences.
 - **Gene flow:**
 - ❑ Gene flow, also known as migration, is the movement of genetic material or individuals from one population to another.
 - ❑ Gene flow can occur when Individuals migrate and reproduce in a new population.
 - (ii) Environmental Variation:**
 - ❑ This type of variation describes differences between organisms caused by exposure to different external conditions.
 - ❑ Environmental variations can be caused by a single factor or multiple factors, such as climate, diet, culture, lifestyle, accidents, temperature, light, moisture and minerals.

➤ **Variation in Asexual Reproduction**

- ✓ An individual organism divides to produce offspring, all of which are genetically identical to the parent, with minor genetic changes due to DNA Copying Errors.
- ✓ **DNA Copying error** are small inaccuracies during DNA replication result in mutations (changes in the genetic material that can introduce slight variations in the offspring).
- ✓ For example, if a single bacterium divides and the resulting two bacteria divide again, producing four bacteria, these would be genetically very similar, with only tiny differences caused by errors in DNA replication.
- ✓ Asexually reproducing organisms (like a field of sugarcane) show very little variation among individual plants.

➤ **Variation in Sexual Reproduction**

- ✓ Sexual reproduction generates much greater diversity compared to asexual reproduction.
- ✓ It is mainly caused by Genetic Recombination.
- ✓ The recombination of genes during sexual reproduction increases the likelihood of producing individuals with unique genetic traits, contributing to the genetic variation in a population.
- ✓ Sexually reproducing organisms (like **human beings** and animals) show distinct and noticeable variations among different individuals.
- ✓ In human populations, earlobes show two distinct variants: **free earlobes** and **attached earlobes**. The lowest part of the ear is closely attached to the side of the head in some individuals, while it is free in others.

➤ **Survival of Variations in the Environment**

- ✓ Variations do not have equal chances of surviving in the environment; their survival depends on how they align with environmental conditions.
- ✓ For example, bacteria with the ability to withstand heat would have a survival advantage during a heat wave. Similarly, other variations could give individuals an edge in different environmental conditions, such as resistance to cold, drought, or disease.
- ✓ **Selection of Variants:** Environmental factors select which variants survive, leading to differential survival and reproduction. This process is known as **Natural selection**.

➤ **Chromosomes:**

- ☐ Chromosomes are thread-like structures located inside the nucleus of animal and plant cells.
- ☐ Each chromosome is made of protein and a single molecule of deoxyribonucleic acid (DNA).

➤ **DNA:**

- ✓ Deoxyribonucleic acid (DNA) is a molecule that contains genetic information for the development, growth, and reproduction of organisms.
- ✓ DNA contains instructions for making proteins.

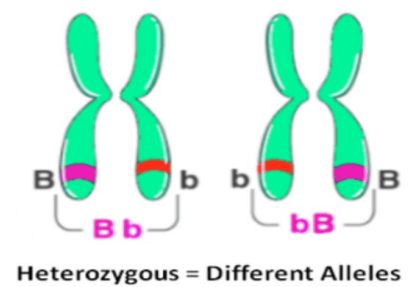
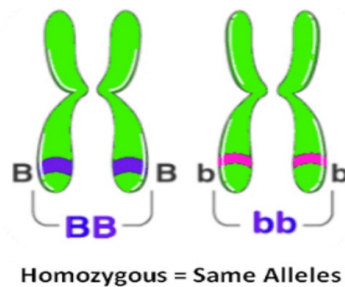
➤ **Gene:**

- ✓ Each section of DNA that encodes the instructions for making a specific protein is called a gene for that protein.
- ✓ Gene is fundamental unit of heredity.
- ✓ It passes from parents to offsprings.
- ✓ Genes are located on chromosomes in the nucleus of cells.
- ✓ Genes carry information that determines the traits of a person.

➤ **Allele:**

- ✓ An allele is one of two or more versions of DNA sequence that can lead to variations in traits.
- ✓ They are located at the same position on a chromosome.

- ✓ An individual inherits two alleles, one from each parent, for any given genomic location where such variation exists.
- ✓ If the two alleles for a trait are the same, the individual is homozygous for that Allele.
- ✓ If the two alleles for a trait are different, the individual is heterozygous for that Allele.



- ✓ **Dominant Alleles/Trait:**

- ☐ An allele of a gene is said to be dominant when it effectively overrules the other allele
 - ☐ It expresses its trait even if only one copy of it is present in the genetic makeup of the organism.
 - ☐ Dominant alleles are usually represented by a capital letter (e.g., "A").
 - ☐ Homozygous dominant is a genotype which has two copies of the same dominant allele.

- ✓ **Recessive Alleles/Traits:**

- ☐ A recessive allele is the allele which does not express itself when it is paired with dominant allele.
 - ☐ Its effect is only observable when two copies of it are present in the genotype.
 - ☐ A recessive allele is masked in the presence of a dominant allele and only expresses its trait when two copies are present.
 - ☐ Recessive alleles are typically represented by a **lowercase letter** (e.g., "a").
 - ☐ Homozygous recessive is a gene which has two copies of the same recessive allele.

- **How Proteins Control Characteristics:**

- ☐ Proteins play a crucial role in determining the characteristics (traits) of an organism. For example, consider plant height. Plant height is influenced by hormones that trigger growth.
 - ☐ The amount of a specific hormone made by the plant determines its height. This hormone is produced through a process that involves enzymes.
 - ☐ If the enzyme responsible for producing the hormone works efficiently, a lot of the hormone will be made, leading to a tall plant.
 - ☐ If the gene that codes for this enzyme has a mutation that makes it less efficient, less hormone will be produced, and the plant will be shorter.
 - ☐ Therefore, genes directly control traits by affecting the efficiency of proteins (like enzymes) that regulate processes within the organism.

- **Germ Cells and Single Gene Set:**

- ☐ In heredity, each germ cell (sperm or egg) must carry only one set of genes, not two.
 - ☐ In body cells, there are two copies of each chromosome (one from each parent). But in germ cells, only one chromosome from each pair is present.
 - ☐ Each germ cell must contain only half the number of chromosomes found in regular body cells. This is achieved through a special type of cell division called meiosis.
 - ☐ When two germ cells (sperm and egg) combine during fertilization, they restore the full number of chromosomes in the offspring, ensuring the genetic stability of the species.

- **Genotype and Phenotype:**

- ✓ **Genotype:**

- ☐ Genotype refers to the entire genetic makeup of an individual.
 - ☐ It represents the two alleles present at a specific location in the gene.

- ✓ **Phenotype**

- ☐ It refers to the observable traits or characteristics of an organism, such as its appearance, behaviour, or physiological traits.
 - ☐ The phenotype is influenced by the organism's genotype and environmental factors.

- ✓ **Example of Genotype and Phenotype:** Consider a gene for flower color in plants, where:
 "A" = dominant allele for red flowers and "a" = recessive allele for white flowers
 Possible Genotypes:
 - (i) AA: Homozygous dominant (two "A" alleles)
 Phenotype: Red flowers (since the dominant allele "A" is expressed in both allele).
 - (ii) Aa: Heterozygous (one "A" and one "a" allele)
 Phenotype: Red flowers (because the dominant "A" allele is expressed, masking the effect of "a").
 - (iii) aa: Homozygous recessive (two "a" alleles)
 Phenotype: White flowers (the recessive trait "a" is expressed because alleles are recessive).

➤ **Mendel's Contribution in Genetics:**

- ✓ The principles of trait inheritance in humans are based on the fact that both the mother and father provide nearly equal amounts of genetic material to their child.
- ✓ Each trait of child is influenced by both the DNA from the father and the DNA from the mother, so a child inherits two versions (alleles) of each trait.
- ✓ **Gregor Mendel** is known as the **Father of modern Genetics**.
- ✓ **Mendel** showed how traits are inherited in child from parents through his experiments on pea plants.
- ✓ He made three basic laws of inheritance
 - (i) The Law of Dominance
 - (ii) The Law of Segregation
 - (iii) The Law of Independent Assortment.
- ✓ **Mendel** crossed plants with different traits (for example, a tall plant with a short plant), observed the offspring, and calculated the percentages of traits in the progeny (next generation).
- ✓ **Reasons to choose Pea Plant for Mendel's Experiment:**
 - (i) **Easy to grow:** Pea plants are easy to cultivate and can be grown in pots or on the ground.
 - (ii) **Short life cycle:** Pea plant is an annual plant with short life cycle of 2-3 months, so large number of offsprings can be analysed within a short period of time.
 - (iii) **Visible characteristics:** Pea plants have several easily recognizable traits, such as round/wrinkled seeds, tall/short plants, and white/violet flowers, to study inheritance.
 - (iv) **Both Self and cross pollinating:** Pea plants are bisexual and can also perform cross pollination.

➤ **Mendel's Experiments:**

- ✓ Gregor Mendel's experiments involved self and cross-pollination of pea plants to study how traits are passed down through generations.
- ✓ His experiments took place between 1856 and 1863.
- ✓ Mendel studied seven different traits in pea plants, including:
 - (i) Pea shape (round or wrinkled)
 - (ii) Pea color (green or yellow)
 - (iii) Plant size (tall or dwarf)
 - (iv) Flower color (purple or white)
 - (v) Position of flowers (axial or terminal)
 - (vi) Pod shape (constricted or inflated)
 - (vii) Pod color (green or yellow)

- ✓ Two experiments of Mendel are stated below:

1) Monohybrid cross:

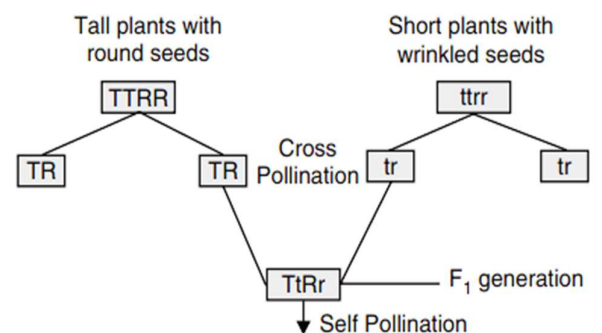
- ✚ A monohybrid cross is a breeding experiment that involves crossing two organisms that differ in only one trait.
- ✚ Steps involved in Monohybrid cross experiment:
 - (i) Mendel crossed two purebred pea plants that differed in a single trait, such as height, flower color, or seed shape. For example, he would cross a pure tall plant with a pure short plant.
 - (ii) The offspring of the cross, called F₁ progeny, resembled the parent with the dominant trait.
 - (iii) Mendel then self-pollinated the F₁ plants with each other to produce the F₂ generation. In the F₂ generation, Mendel observed a 3:1 ratio of plants with the dominant trait to plants with the recessive trait.

phenotypes of parents	tall		dwarf										
genotypes of parents	TT		×	tt									
gametes	(T)	(T)	×	(t) (t)									
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F ₁ genotypes	all Tt												
F ₁ phenotypes	all tall												

phenotypes of parents	tall		dwarf									
genotypes of parents	Tt	×	Tt									
gametes	(T) (t)	×	(T) (t)									
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F ₁ genotypes	1 TT, 2 Tt, 1 tt											
F ₁ phenotypes	tall tall dwarf											
ratio	3 tall : 1 dwarf											

2) Dihybrid cross:

- ✚ A Dihybrid cross is a breeding experiment that involves crossing two organisms that differ in two traits.
- ✚ Tallness and round seeds are dominant traits, while shortness and wrinkled seeds are recessive traits.
- ✚ Steps involved in Dihybrid cross experiment:
 - (i) Mendel crossed two purebred pea plants that differed in a two trait. For example, he would cross a pure tall plant with round seeds and pure short plants with wrinkled seeds.
 - (ii) The offspring of the cross, F₁ progeny will all be tall with round seeds.
 - (iii) When F₁ plants self-pollinated with each other to produce the F₂ progeny, observations are:
 - (iv) 9 Tall with Round seeds
 - (v) 3 Tall with Wrinkled seeds
 - (vi) 3 Short with Round seeds
 - (vii) 1 Short with Wrinkled seeds



♀ \ ♂	TR	Tr	tR	tr
TR	TTRR Tall Round	TTRr Tall Round	TtRR Tall Round	TtRr Tall Round
Tr	TTRr Tall Round	TTrr Tall wrinkled	TtRr Tall Round	Tttr Tall wrinkled
tR	TtRR Tall Round	TtRr Tall Round	ttRR Short Round	ttRr Short Round
tr	TtRr Tall Round	Tttr Tall wrinkled	ttRr Short Round	tttr Short wrinkled

- (iv) The phenotype ratio will be 9:3:3:1
- (v) The Genotype ratio result is 1:2:2: 4:1:2:1:2:1

✓ **Conclusion of Mendel's Experiment:**

- ✚ According to Mendelian genetics, both parents contribute equally to the DNA of their offspring during sexual reproduction.
- ✚ Based on his experiments, he observed that traits which were absent in the F1 generation had reappeared in the F2 generation.
- ✚ Based on such fact, Mendel has given three important Laws of Inheritance:

(i) **Law of dominance:** Some alleles are dominant over others, meaning that if an organism has at least one dominant allele, it will display the dominant trait.

(ii) **Law of segregation:** During the formation of gametes, the two alleles for each gene separate so that each offspring inherits one allele from each parent.

(iii) **Law of independent assortment:**

- ❏ During formation of gametes, alleles of different genes segregate independently and are distributed independently in the next generation.
- ❏ Each gene set is present **not as a single long thread of DNA, but as separate independent pieces called chromosomes**. Each cell has two copies of each chromosome (one maternal and one paternal). Germ cells take one chromosome from each pair, allowing separate traits to be inherited independently.
- ❏ Each cell contains two copies of each chromosome (one from each parent), but during meiosis, each germ cell gets only one chromosome from each pair. These chromosomes may come from either the maternal or paternal side.
- ❏ If both parents contribute a single chromosome from each pair, it allows for the independent inheritance of traits, as shown in Mendel's experiments. If genes were linked on the same chromosome, they would not assort independently.

➤ **Sex Determination:**

✓ **Sex Determination in Different Species:**

- ❏ Various species have different strategies for determining sex.
- ❏ In some species, the sex of an individual is determined by environmental factors, rather than genetics. For example, in certain reptiles, such as some turtles and crocodiles, the temperature at which eggs are incubated can decide whether the offspring will be male or female.
- ❏ Some species, such as certain fish and snails, can change sex during their lifetime. These species may start life as one sex and later switch to the opposite sex in response to environmental factors or social conditions, a phenomenon known as sequential hermaphroditism. This shows that sex is not always genetically determined.

✓ **Human Sex Determination:**

- ❏ In humans, sex is largely determined by genetic factors.
- ❏ The sex of an individual is decided by the genes inherited from their parents, specifically the sex chromosomes.

✓ **Chromosomal Pairs in Humans:**

- ❏ Humans have 46 chromosomes in total, organized into 23 pairs.
- ❏ Of the 23 pairs, 22 are autosomes (non-sex), which are the same in both males and females.
- ❏ The 23rd pair consists of sex chromosomes, which determine an individual's biological sex.
- ❏ One chromosome in each pair is inherited from the mother, and the other from the father

✓ **Sex Chromosomes in Women and Men:**

- ❑ Women have two X chromosomes (XX), making their sex chromosome pair identical.
- ❑ Men have one X chromosome and one Y chromosome (XY), so their sex chromosome pair is different.

✓ **Inheritance of Sex Chromosomes:**

- ❑ All children inherit one X chromosome from their mother, regardless of sex.
- ❑ The father's contribution determines the child's sex:
- ❑ If the father passes on an X chromosome, the child will be female (XX).
- ❑ If the father passes on a Y chromosome, the child will be male (XY).
- ❑ The child's sex is determined by the sex chromosome inherited from the father: X for girls and Y for boys.

